

Code-:

```
DROP DATABASE IF EXISTS bookflow_db;  
CREATE DATABASE bookflow_db;  
USE bookflow_db;
```

```
CREATE TABLE Books (  
    book_id INT PRIMARY KEY,  
    title VARCHAR(100) NOT NULL,  
    isbn VARCHAR(20) UNIQUE,  
    published_year INT CHECK (published_year < 2027)  
);
```

```
INSERT INTO Books (book_id, title, isbn, published_year) VALUES  
(1, 'The Great Gatsby', '9780743273565', 1925),  
(2, 'To Kill a Mockingbird', '9780061120084', 1960),  
(3, '1984', '9780451524935', 1949);
```

```
CREATE TABLE Members (  
    member_id INT PRIMARY KEY,  
    full_name VARCHAR(100),  
    email VARCHAR(100) UNIQUE  
);
```

```
INSERT INTO Members (member_id, full_name, email) VALUES  
(101, 'John Smith', 'john.smith@email.com'),  
(102, 'Emma Wilson', 'emma.wilson@email.com'),  
(103, 'Michael Brown', 'michael.brown@email.com');
```

```
CREATE TABLE Loans (  
    loan_id INT PRIMARY KEY,  
    member_id INT,  
    book_id INT,  
    loan_date DATE,  
    FOREIGN KEY (member_id) REFERENCES Members(member_id),  
    FOREIGN KEY (book_id) REFERENCES Books(book_id)  
);
```

```
INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES  
(1, 101, 1, '2025-01-05'),  
(2, 102, 2, '2025-01-08'),  
(3, 103, 3, '2025-01-10'),  
(4, 101, 2, '2025-02-01'),
```

```
(5, 102, 1, '2025-02-05'),  
(6, 103, 2, '2025-02-12'),  
(7, 101, 3, '2025-03-01'),  
(8, 102, 3, '2025-03-07'),  
(9, 103, 1, '2025-03-15'),  
(10, 101, 1, '2025-04-01');
```

```
SELECT m.full_name AS Member_Name, b.title AS Book_Title  
FROM Loans l  
INNER JOIN Members m ON l.member_id = m.member_id  
INNER JOIN Books b ON l.book_id = b.book_id;
```

```
SELECT published_year, COUNT(book_id) AS Total_Books  
FROM Books  
GROUP BY published_year  
ORDER BY published_year;
```

```
CREATE TABLE Donation_History (  
    donation_id INT PRIMARY KEY,  
    book_id INT,  
    donor_name VARCHAR(100),  
    donation_date DATE,  
    FOREIGN KEY (book_id) REFERENCES Books(book_id)  
);
```

```
START TRANSACTION;
```

```
INSERT INTO Books (book_id, title, isbn, published_year)  
VALUES (4, 'Animal Farm', '9780451526342', 1945);
```

```
INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)  
VALUES (1, 4, 'Raj Kumar', CURDATE());
```

```
COMMIT;
```

```
CREATE INDEX idx_books_isbn ON Books(isbn);
```

```
SELECT * FROM Books WHERE isbn = '9780451524935';
```

Output:-

```
[mysql> SHOW TABLES;
```

```
+-----+
| Tables_in_bookflow_db |
+-----+
| Books                  |
| Donation_History      |
| Loans                  |
| Members                |
+-----+
4 rows in set (0.001 sec)
```

```
[mysql> SELECT * FROM Books;
```

```
+-----+-----+-----+-----+
| book_id | title                | isbn          | published_year |
+-----+-----+-----+-----+
| 1       | The Great Gatsby    | 9780743273565 | 1925           |
| 2       | To Kill a Mockingbird | 9780061120084 | 1960           |
| 3       | 1984                 | 9780451524935 | 1949           |
| 4       | Animal Farm         | 9780451526342 | 1945           |
+-----+-----+-----+-----+
4 rows in set (0.000 sec)
```

```
[mysql> SELECT * FROM Donation_History;
```

```
+-----+-----+-----+-----+
| donation_id | book_id | donor_name | donation_date |
+-----+-----+-----+-----+
| 1           | 4       | Raj Kumar  | 2026-07-29    |
+-----+-----+-----+-----+
1 row in set (0.001 sec)
```

```
[mysql> SELECT * FROM Loans;
```

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01

```
10 rows in set (0.001 sec)
```

```
[mysql> SELECT * FROM Members;
```

member_id	full_name	email
101	John Smith	john.smith@email.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com

```
3 rows in set (0.000 sec)
```